

Terms Rewrite Systems Equational Reasoning, Complete Property

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Abstract. Term rewriting systems (TRSs) provide a computational model for equational reasoning, where equations are oriented into directed rules and applied as local transformations. A central question in this setting is whether a TRS is complete with respect to an underlying equational theory — that is, whether rewriting alone is sufficient to decide equational validity. This paper surveys the main notions and methods surrounding this property. We first recall the standard framework of TRSs and equational theories, then examine the conditions under which completeness holds, with particular attention to how it can be preserved under disjoint unions of systems. We review classical results connecting completeness to confluence and termination, discuss criteria for detecting its failure, and present the principal techniques for restoring it — including syntactic unification and Knuth–Bendix completion. This survey aims to give a unified and accessible account of completeness in term rewriting, serving as a reference for readers interested in the foundations of equational logic and automated deduction.

Keywords. term rewriting, confluence, termination, normalisation, complete, unification, modularity, Knuth–Bendix completion, equational reasoning.

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1 Introduction

Equational reasoning constitutes one of the foundational pillars of mathematical logic and theoretical computer science, providing a uniform framework for describing the algebraic properties of computational structures. Term Rewriting Systems (TRS) offer a particularly elegant and powerful approach to this problem, by orienting equational axioms into directed rewrite rules, thereby transforming the undirected question of equational provability into a deterministic computation of normal forms. Central to this framework is the notion of confluence — the property ensuring that any two reduction sequences issuing from a common term eventually converge to a single result. Far from being a mere technical convenience, confluence is the cornerstone upon which the entire decision-theoretic apparatus of equational reasoning rests: it guarantees the uniqueness of normal forms, ensures the coherence of the induced equational theory, and, when combined with termination, yields the decidability of the word problem.

Our guiding goal throughout this paper is *equational reasoning itself*: given a set of equational axioms, we want an effective, mechanical procedure to decide whether two terms are provably equal. Term rewriting is the vehicle for this goal, and confluence together with termination is exactly what makes the vehicle run: it is what turns the static congruence $=_E$ into a computable equivalence, decidable by comparing normal forms (Section 3). Everything else in this paper is in the service of that single idea. We first develop the classical machinery for establishing confluence — Newman’s Lemma, syntactic unification, and the Critical Pair Theorem (Sections 3 to 5) — and the constructive procedure that restores confluence when it fails, Knuth–Bendix completion (Section 6). We then turn to a more peripheral, though instructive, question: how these properties behave under disjoint unions of systems (Section 7), where confluence turns out to be modular but termination is not. We close with a fundamental limit on the whole programme: termination of an arbitrary TRS is itself undecidable (Section 8).

Reading this paper. Section 2 lays down the syntactic apparatus (terms, substitutions, equational theories, positions, rewrite steps). Section 3 develops confluence, termination, and their interaction via Newman’s Lemma — the technical heart of the equational-reasoning programme. Section 4 develops syntactic unification, a prerequisite for the algorithmic construction of critical pairs, presented immediately afterwards in Section 5 together with the Critical Pair Theorem. Section 6 presents the Knuth–Bendix completion procedure, which effectively restores confluence using critical pairs, and its correctness theorem — at this point, the central goal of the paper (an effective decision procedure for equational validity) is fully achieved. Section 7 is a secondary study of how confluence, termination, and convergence behave under disjoint unions of systems; it is not required for the main line of the paper and may be skipped on a first reading. Section 8 is a self-contained appendix-like section showing, via an encoding of Turing machines, that termination of an arbitrary TRS is undecidable.

2 Preliminaries

2.1 Terms and signatures

Definition (*signature*). A *signature* Σ is a set of function symbols, each equipped with an *arity* $\text{ar} : \Sigma \rightarrow \mathbb{N}$. Symbols of arity 0 are called *constants*.

Nota. We fix two denumerable sets $\mathcal{X}$ and $\mathcal{C}$, of *variables* and *constants* respectively, disjoint from Σ and from each other.

Definition (Terms). A *term* over Σ and a set of atoms $\mathcal{A}$ (either $\mathcal{X}$ or $\mathcal{C}$) is defined inductively: every element $a \in \mathcal{A}$ is a term, and if $f \in \Sigma$ with $\text{ar}(f) = n$ and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term.

Note. We write $\mathcal{T}(\Sigma, \mathcal{X})$ for the set of *patterns* (terms built over the variables $\mathcal{X}$), and $\mathcal{T}(\Sigma, \mathcal{C})$ for the set of *ground terms* (terms built over the constants $\mathcal{C}$, containing no variable).

Example. Let $\Sigma = \{0, s, +\}$ with arities 0, 1, 2 respectively (0 a constant, s a unary successor symbol, $+$ a binary addition symbol). Let $\mathcal{C} = \{a, b, c, \dots\}$ and $\mathcal{X} = \{x, y, z, \dots\}$.

- *Ground terms* ($\mathcal{T}(\Sigma, \mathcal{C})$, no variable): 0 , $s(0)$, $s(s(0))$, $+(s(0), s(s(0)))$, and also $+(a, s(b))$ once $a, b \in \mathcal{C}$ are used as free constants.
- *Patterns* ($\mathcal{T}(\Sigma, \mathcal{X})$, possibly containing variables): $+(x, s(0))$, $s(s(y))$, $+(x, +(y, z))$.

Ground terms are the terms with no variable occurrence; patterns are the general syntactic objects manipulated by rewrite rules, whose variables get instantiated by substitution.

Definition (Substitution). A *substitution* σ is a map $\sigma : \mathcal{X} \rightarrow \mathcal{T}(\Sigma, \mathcal{X})$.

Note. Generally, we want σ to have finite support, i.e.

$$|\{x \in \mathcal{X} : \sigma(x) \neq x\}| < +\infty.$$

Note (extension). A substitution does not apply only to variables, but to all terms, by structural extension:

- if x is a variable: $x\sigma = \sigma(x)$;
- if $f(t_1, \dots, t_n)$ is a term: $f(t_1, \dots, t_n)\sigma = f(t_1\sigma, \dots, t_n\sigma)$.

2.2 Equational theories

Definition (Congruence). Let A be a set equipped with a family of operations $(\omega_i)_{i \in I}$, each $\omega_i : A^{n_i} \rightarrow A$ of arity n_i (this is called an *algebra*, or Ω -*algebra*, in the sense of universal algebra; groups, rings, and term algebras are all instances of this notion, each with their own choice of operations ω_i — for a group, $\Omega = \{., ^{-1}, e\}$; for a term algebra over Σ , $\Omega = \Sigma$ itself, each symbol $f \in \Sigma$ acting as an operation of arity $\text{ar}(f)$). A *congruence* on A is an equivalence relation $\sim$ on A that is *compatible* with every operation: for every $i \in I$ and every $a_1, \dots, a_{n_i}, b_1, \dots, b_{n_i} \in A$,

$$a_1 \sim b_1, \dots, a_{n_i} \sim b_{n_i} \implies \omega_i(a_1, \dots, a_{n_i}) \sim \omega_i(b_1, \dots, b_{n_i}).$$

Definition (Quotient set induced by a congruence). Given a congruence $\sim$ on A , the *quotient set* $A/\sim$ is the set of equivalence classes $[a]_\sim = \{b \in A : a \sim b\}$ for $a \in A$. Each operation ω_i descends to a well-defined operation on $A/\sim$, by

$$\omega_i([a_1]_\sim, \dots, [a_{n_i}]_\sim) := [\omega_i(a_1, \dots, a_{n_i})]_\sim,$$

this definition being independent of the choice of representatives $a_1, \dots, a_{n_i}$ precisely because $\sim$ is a congruence (this is exactly what compatibility guarantees).

Note. Specializing to $A = \mathcal{T}(\Sigma, \mathcal{X})$ with operations given by Σ itself (each $f \in \Sigma$ acting as an operation of arity $\text{ar}(f)$), a congruence on terms is exactly an equivalence relation $\sim$ such that $t_i \sim v_i$ for every i implies $f(t_1, \dots, t_n) \sim f(v_1, \dots, v_n)$ for every $f \in \Sigma$. This is the notion of congruence used throughout this paper.

Definition (*Equational theory*). An *equational theory* E over a signature Σ is a set of equations $s \approx t$ with $s, t \in \mathcal{T}(\Sigma, \mathcal{X})$.

Definition (*Equational congruence induced by E*). Given an equational theory E , we write $=_E$ for the smallest congruence on $\mathcal{T}(\Sigma, \mathcal{X})$, stable under substitution, containing every instance $s\sigma \approx t\sigma$ for $s \approx t \in E$ and every substitution σ .

Note. Every equational theory E can be read as a term rewriting system, by orienting each equation $s \approx t$ into a directed rule pointing from one side to the other — this is precisely the purpose of term rewriting, introduced in the next section: to turn the undirected relation $=_E$ into a computable, directed process.

2.3 Rewrite rules

Definition (*Term Rewriting System (TRS)*). A *TRS* is a set R of rules $l \rightarrow_R r$ where $l, r \in \mathcal{T}(\Sigma, \mathcal{X})$, and we say that l *rewrites to* r . We also write (Σ, R) to say a set of rules on the alphabet or signature Σ .

Definition (*Position and subterm at a position*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$. We define simultaneously, by induction on the structure of t , the set $\text{Pos}(t) \subseteq (\mathbb{N}_{>0})^*$ of *positions* of t (finite sequences of positive integers), and, for every $p \in \text{Pos}(t)$, the *subterm of t at position p* , written $t|_p \in \mathcal{T}(\Sigma, \mathcal{X})$:

- If $t \in \mathcal{X} \cup \mathcal{C}$ (a variable or a constant):

$$\text{Pos}(t) = \{\varepsilon\}, \quad t|_\varepsilon = t.$$

- If $t = f(t_1, \dots, t_n)$ with $n \geq 1$:

$$\text{Pos}(t) = \{\varepsilon\} \cup \{i \cdot p : 1 \leq i \leq n, p \in \text{Pos}(t_i)\}, \quad t|_\varepsilon = t, \quad t|_{i \cdot p} = t_i|_p.$$

The empty sequence ε always denotes the *root* of t , with $t|_\varepsilon = t$.

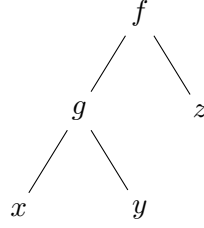
Example. Let $t = f(g(x, y), z)$. Then

$$\text{Pos}(t) = \{\varepsilon, 1, 2, 1 \cdot 1, 1 \cdot 2\},$$

with

$$t|_\varepsilon = f(g(x, y), z), \quad t|_1 = g(x, y), \quad t|_2 = z, \quad t|_{1 \cdot 1} = x, \quad t|_{1 \cdot 2} = y.$$

Position 1 designates the first argument of f (here $g(x, y)$); positions starting with 1 then continue into the arguments of g (hence 1·1 and 1·2); position 2 designates the second argument of f (here z , a variable, hence with no further extension).



Definition (*Replacement at a position*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$, $p \in \text{Pos}(t)$, and $r \in \mathcal{T}(\Sigma, \mathcal{X})$. The term $t[r]_p$ (t with its subterm at position p replaced by r) is defined by induction on p :

$$t[r]_\varepsilon = r, \quad f(t_1, \dots, t_n)[r]_{i.p} = f(t_1, \dots, t_i[r]_p, \dots, t_n).$$

By construction, $(t[r]_p)|_p = r$: replacing at p and then reading back at p recovers exactly r .

Example. With $t = f(g(x, y), z)$ as before, and $r = h(0)$:

$$t[r]_1 = f(h(0), z), \quad t[r]_2 = f(g(x, y), h(0)), \quad t[r]_{1.1} = f(g(h(0), y), z).$$

Nota.

- For $t \in \mathcal{T}(\Sigma, \mathcal{X})$ not a variable, $\text{root}(t)$ denotes the outermost function symbol of t : if $t = f(t_1, \dots, t_n)$, then $\text{root}(t) = f$.
- For p, q two positions, we write $p \leq q$ if p is a *prefix* of q , i.e. there exists p' such that $q = p \cdot p'$; in this case $t|_q = (t|_p)|_{p'}$.
- We write $p < q$ if $p \leq q$ and $p \neq q$ (*strict prefix*): the subterm at q occurs strictly inside the subterm at p .
- We write $p \parallel q$ if p and q are *disjoint* (or *incomparable*), i.e. neither $p \leq q$ nor $q \leq p$. In this case, for any term t with $p, q \in \text{Pos}(t)$ and any term r , we have $(t[r]_p)|_q = t|_q$ (replacing at p does not affect a disjoint position q).
- For two positions $p, q \in (\mathbb{N}_{>0})^*$, exactly one of the following holds:

$$p = q, \quad p < q, \quad q < p, \quad p \parallel q.$$

Note. Terms are naturally pictured as ordered trees: a leaf for each variable or constant, and a node labeled f with n ordered children for each subterm $f(t_1, \dots, t_n)$ — the position $p \in \text{Pos}(t)$ then designates a node of this tree, and $t|_p$ the subtree rooted there. This picture is purely informal and is never used directly in proofs, which always argue on positions and substitutions. In a similar spirit, we freely use *triangle* (or *diamond*) *diagrams* throughout this paper — e.g. a peak $t \xrightarrow{R} s \rightarrow_R u$ joined by a common reduct w — as an informal visual aid alongside the formal argument; these diagrams carry no formal content beyond what is proved in the surrounding text.

Definition (*Redex*). Let R be a TRS, $t \in \mathcal{T}(\Sigma, \mathcal{X})$, and $p \in \text{Pos}(t)$. We say $t|_p$ is a *redex* (of R) if there exist a rule $l \rightarrow r \in R$ and a substitution σ such that $t|_p = l\sigma$.

Definition (Rewrite step). Let R be a TRS, $t \in \mathcal{T}(\Sigma, \mathcal{X})$, $p \in \text{Pos}(t)$ such that $t|_p$ is a redex via rule $\ell \rightarrow r \in R$ and substitution σ . The *rewrite step at position p using $\ell \rightarrow r$* replaces $t|_p$ by $r\sigma$ inside t :

$$t \xrightarrow[p]{\ell \rightarrow r} t[r\sigma]_p.$$

We write $t \rightarrow_R t'$ if $t' = t[r\sigma]_p$ for some rule $\ell \rightarrow r \in R$, position $p \in \text{Pos}(t)$, and substitution σ with $t|_p = \ell\sigma$.

Example. Let $\Sigma = \{0, s, +\}$ and $R = \{+(0, y) \rightarrow y, +(s(x), y) \rightarrow s(+ (x, y))\}$. Consider $t = +(s(0), s(0))$.

- At $p = \varepsilon$: $t|_\varepsilon = t = +(s(0), s(0))$ is a redex via the second rule, with $\sigma(x) = 0$, $\sigma(y) = s(0)$. So $t \xrightarrow[\varepsilon]{+(s(x), y) \rightarrow s(+ (x, y))} s(+ (0, s(0)))$.
- In the resulting term $s(+ (0, s(0)))$, the subterm at position 1 is $+ (0, s(0))$, a redex via the first rule with $\sigma(y) = s(0)$: $s(+ (0, s(0))) \xrightarrow[1]{+(0, y) \rightarrow y} s(s(0))$.

So $t \rightarrow_R^+ s(s(0))$ (in two steps), and $s(s(0))$ is a normal form: no subterm of $s(s(0))$ matches a left-hand side of R .

Definition (Iterated rewriting). Let R be a TRS and $a, b \in \mathcal{T}(\Sigma, \mathcal{X})$. For $n \in \mathbb{N}$, we define $a \rightarrow_R^n b$ ("a rewrites to b in exactly n steps") by induction on n :

$$a \rightarrow_R^0 b \iff a = b, \quad a \rightarrow_R^{n+1} b \iff \exists c, a \rightarrow_R c \text{ and } c \rightarrow_R^n b.$$

We further write

$$a \rightarrow_R^* b \iff \exists n \in \mathbb{N}, a \rightarrow_R^n b, \quad a \rightarrow_R^+ b \iff \exists n \in \mathbb{N}^*, a \rightarrow_R^n b.$$

Definition (Conversion). Let R be a TRS. We write $a \leftrightarrow_R b$ if $a \rightarrow_R b$ or $b \rightarrow_R a$. For $n \in \mathbb{N}$, $a \leftrightarrow_R^n b$ is defined by induction exactly as above, with $\leftrightarrow_R$ in place of $\rightarrow_R$. We write $a \leftrightarrow_R^* b$ if $\exists n, a \leftrightarrow_R^n b$: this is the smallest equivalence relation containing $\rightarrow_R$, called *conversion*.

Note. $\leftrightarrow_R^*$ is exactly the smallest congruence containing $\rightarrow_R$ and stable under substitution and context — it coincides with $=_E$ when R is obtained by orienting every equation of E (in either direction).

Definition (Joinable). Two terms a and b are joinable if there exists a term c such that $a \rightarrow_R^* c$ and $b \rightarrow_R^* c$; we write it $a \downarrow_R b$.

Definition (Pre-diamond). If a, b, c are three terms such that $b \leftarrow a \rightarrow c$, the triple (a, b, c) is called a *pre-diamond*, and a is its *root*. A pre-diamond is *proper* if a, b, c are pairwise distinct. A pre-diamond (a, b, c) is *joinable* if the pair (b, c) is joinable.

Note. A non-proper pre-diamond is always joinable.

Definition (normal form). Let R be a term rewriting system over $\mathcal{T}(\Sigma, \mathcal{X})$. A term t is said to be a *normal form* (or *irreducible*) for R if there is no term t' such that $t \rightarrow_R t'$.

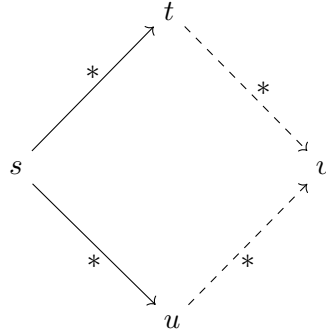
Definition (*Term with a normal form*). Let R be a TRS and $u \in \mathcal{T}(\Sigma, \mathcal{X})$. We say that u has a normal form if there exists a normal form $t \in \mathcal{T}(\Sigma, \mathcal{X})$ for R such that $u \leftrightarrow_R^* t$.

Definition (*Weak normalization*). A TRS R is *weakly normalizing* (WN) if every term $t \in \mathcal{T}(\Sigma, \mathcal{X})$ has a normal form.

3 Core Properties

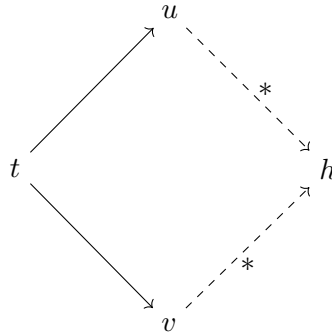
3.1 Confluence and Termination

Definition (*Confluence*). A TRS R is *confluent* if, for all terms $s, t, u \in \mathcal{T}(\Sigma, \mathcal{X})$ with $s \rightarrow_R^* t$ and $s \rightarrow_R^* u$, there exists v such that $t \rightarrow_R^* v$ and $u \rightarrow_R^* v$.



Remark. This definition can be read through the lens of equational theories: if R orients every equation of an equational theory E , confluence of R is precisely what guarantees that joinability by $\rightarrow_R^*$ coincides with (and thus effectively computes) equality in $=_E$ — see the discussion of the word problem below.

Definition (*Local confluence*). A TRS R is *locally confluent* if for all $t, u, v \in \mathcal{T}(\Sigma, \mathcal{X})$ with $t \rightarrow_R u$ and $t \rightarrow_R v$, there exists h such that $u \rightarrow_R^* h$ and $v \rightarrow_R^* h$.



Equivalently, a TRS R is locally confluent iff every pre-diamond is joinable: the only difference with confluence above is that the two initial arrows are single rewrite steps rather than arbitrary reductions.

Definition (*Church–Rosser property*). A TRS R is called *Church–Rosser* if for all $a, b \in \mathcal{T}(\Sigma, \mathcal{X})$ such that $a \leftrightarrow_R^* b$, there exists v such that $a \rightarrow_R^* v$ and $b \rightarrow_R^* v$.

Definition (*Termination*). $\rightarrow_R$ is *terminating* if every rewrite sequence $t_0 \rightarrow_R t_1 \rightarrow_R t_2 \rightarrow_R \dots$ is finite.

Definition (*Noetherian order*). A strict ordering is *Noetherian* (or *well-founded*) if every non-empty set has a minimal element.

Definition (*Ordering relation associated with a TRS*). Given a terminating TRS R on $\mathcal{T}(\Sigma, \mathcal{X})$, we denote $\prec_R$ the relation:

$$\forall a, b \in \mathcal{T}(\Sigma, \mathcal{X}), \quad b \prec_R a \iff a \rightarrow_R^+ b.$$

Proposition (*Properties of $\prec_R$*).

1. Since $\rightarrow_R^+$ is the transitive closure of $\rightarrow_R$, it is trivial that $\prec_R$ is a strict pre-order.
2. Since R terminates, $\prec_R$ is furthermore antisymmetric, and thus a strict ordering relation.
3. Again since R terminates, $\prec_R$ is a Noetherian ordering.

Proof. 1. $\prec_R$ is transitive. Let $a, b, c \in \mathcal{T}(\Sigma, \mathcal{X})$ with $c \prec_R b$ and $b \prec_R a$, i.e. $b \rightarrow_R^+ c$ and $a \rightarrow_R^+ b$. Since $\rightarrow_R^+$ is the transitive closure of $\rightarrow_R$, $a \rightarrow_R^+ b$ and $b \rightarrow_R^+ c$ give $a \rightarrow_R^+ c$, i.e. $c \prec_R a$. Hence $\prec_R$ is transitive; it is a strict pre-order (irreflexivity being immediate, since R terminates and $a \rightarrow_R^+ a$ would give an infinite reduction $a \rightarrow_R a \rightarrow_R a \rightarrow_R \dots$).

2. $\prec_R$ is antisymmetric. Suppose, for contradiction, that $a \prec_R b$ and $b \prec_R a$ for some $a \neq b$. This means $b \rightarrow_R^+ a$ and $a \rightarrow_R^+ b$. Concatenating a witnessing sequence of steps for each, we obtain an infinite reduction $a \rightarrow_R \dots \rightarrow_R b \rightarrow_R \dots \rightarrow_R a \rightarrow_R \dots$, contradicting termination of R . Hence $\prec_R$ is antisymmetric, and therefore a strict ordering relation.

3. $\prec_R$ is Noetherian. Suppose, for contradiction, that some non-empty set $A \subseteq \mathcal{T}(\Sigma, \mathcal{X})$ has no minimal element for $\prec_R$. Fix $t_0 \in A$. Since t_0 is not minimal in A , there exists $t_1 \in A$ with $t_1 \prec_R t_0$. By induction, having built $t_0, \dots, t_n \in A$ with $t_n \prec_R \dots \prec_R t_0$, since t_n is not minimal in A there exists $t_{n+1} \in A$ with $t_{n+1} \prec_R t_n$. This produces an infinite sequence $t_0 \succ_R t_1 \succ_R t_2 \succ_R \dots$ in A . By definition of $\prec_R$, each $t_k \succ_R t_{k+1}$ means $t_k \rightarrow_R^+ t_{k+1}$, i.e. there exists a finite non-empty sequence of rewrite steps from t_k to t_{k+1} . Concatenating these sequences for every k yields an infinite reduction sequence issued from t_0 , contradicting termination of R . Hence every non-empty set has a $\prec_R$ -minimal element: $\prec_R$ is Noetherian. $\square$

Definition (*Complete*). R is *complete*, or *convergent*, if R is terminating and confluent.

Remark. If R is convergent and orients an equational theory E (i.e. $\leftrightarrow_R^* = =_E$), then for all $t, t' \in \mathcal{T}(\Sigma, \mathcal{X})$:

$$t =_E t' \iff \exists u, t \rightarrow_R^* u \text{ and } t' \rightarrow_R^* u.$$

Indeed, $t =_E t'$ means $t \leftrightarrow_R^* t'$; since R is confluent, this conversion yields a common

reduct u by joinability. This is precisely what makes the word problem for E decidable once R is convergent: compute the (unique) normal forms of t and t' and compare them syntactically.

Proposition (*Canonical representatives via convergence*). Let E be an equational theory and R a convergent TRS orienting E (i.e. $\leftrightarrow_R^* = =_E$). Then the map sending each equivalence class $[t]_{=E} \in \mathcal{T}(\Sigma, \mathcal{X})/_E$ to the (unique) normal form of any of its representatives is well defined, and gives a bijection

$$\mathcal{T}(\Sigma, \mathcal{X})/_E \xrightarrow{\sim} \{\text{normal forms of } R\}.$$

Proof. Well defined. Let $t, t' \in [u]_{=E}$, i.e. $t \leftrightarrow_R^* t'$. Since R is confluent, t and t' have a common reduct w ; since R terminates, t and t' each reduce to some normal form, and by confluence these normal forms coincide (a normal form cannot reduce further, so w itself must reduce to, and hence equal, the normal form of both t and t'). Hence the normal form does not depend on the choice of representative within the class.

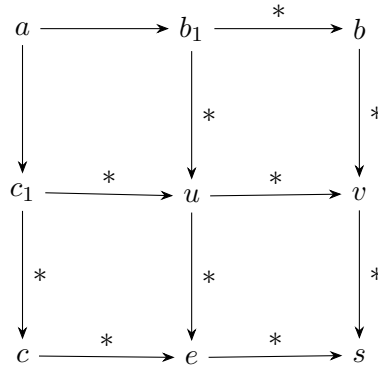
Injective. If two classes have the same normal form $\bar{u}$, then $t \rightarrow_R^* \bar{u} \leftarrow_R^* t'$, so $t \leftrightarrow_R^* t'$: the two classes coincide.

Surjective. Every normal form $\bar{u}$ is trivially the normal form of $[\bar{u}]_{=E}$. $\square$

3.2 Newman's Lemma

Newman's Lemma is the central bridge between the *local* property of confluence (checkable, in principle, redex by redex) and the *global* property of confluence (joinability of arbitrarily long derivations). We give two independent proofs below: the first proceeds by structural induction along the well-founded order $\prec_R$, directly establishing the desired joinability property; the second argues by contradiction on a minimal counterexample. Both are standard in the literature, and we present them side by side because they illustrate two complementary proof techniques (direct well-founded induction versus minimal-counterexample) that recur throughout this paper — in particular, the second technique is exactly the one used later for the Critical Pair Theorem and for the modularity of confluence.

Theorem (*Newman's Lemma*). A *locally confluent* and *terminating* term rewriting system is confluent.



Proof 1 (well-founded induction). Since R is terminating, the relation $\rightarrow_R$ is well-founded, and thus induces the order relation $\prec_R$ of the previous section. The diagram above summarises the construction carried out below: starting from the peak a , local confluence at the

first step produces u , and three successive applications of the induction hypothesis P (at c_1 , at b_1 , and at u) chase the diagram down to a common point s .

The fact that R terminates allows us to carry out a proof by induction to show the property P defined by

$$\forall a \in \mathcal{T}(\Sigma, \mathcal{X}), \quad P(a) : \forall b, c \text{ with } a \rightarrow_R^* b \text{ and } a \rightarrow_R^* c, \exists d, b \rightarrow_R^* d \text{ and } c \rightarrow_R^* d.$$

Base case. The base case concerns terms a that do not rewrite to any other term. Let such an a be given. For b, c such that $a \rightarrow_R^* b$ and $a \rightarrow_R^* c$, we can only have $c = b = a$. So $d = a$ gives us $b \rightarrow_R^* d$ and $c \rightarrow_R^* d$. Hence $P(a)$.

Inductive case. Let a ; assume that for all $a' \prec_R a$, $P(a')$ holds. Let b, c be such that $a \rightarrow_R^* c$ and $a \rightarrow_R^* b$. Write $a \rightarrow_R c_1 \rightarrow_R^* c$ and $a \rightarrow_R b_1 \rightarrow_R^* b$.

By local confluence, there exists u such that $c_1 \rightarrow_R^* u$ and $b_1 \rightarrow_R^* u$.

Since $c_1 \rightarrow_R^* u$ and $c_1 \rightarrow_R^* c$, and $c_1 \prec_R a$, $P(c_1)$ holds: there exists e such that $u \rightarrow_R^* e$ and $c \rightarrow_R^* e$.

Since $b_1 \rightarrow_R^* b$ and $b_1 \rightarrow_R^* u$, and $b_1 \prec_R a$, $P(b_1)$ holds: there exists v such that $b \rightarrow_R^* v$ and $u \rightarrow_R^* v$.

We now have $u \rightarrow_R^* e$ and $u \rightarrow_R^* v$, and $u \prec_R a$, so $P(u)$ holds: there exists s such that $e \rightarrow_R^* s$ and $v \rightarrow_R^* s$. Then $b \rightarrow_R^* v \rightarrow_R^* s$ and $c \rightarrow_R^* e \rightarrow_R^* s$: the peak b, c is joinable, which concludes the proof. $\square$

Proof 2 (minimal counterexample). Let R be a terminating and locally confluent TRS. Let

$$A = \{a \in \mathcal{T}(\Sigma, \mathcal{X}) \mid \exists c, b : c \leftarrow_R^+ a \rightarrow_R^+ b, \nexists d, b \rightarrow_R^* d \leftarrow_R^* c\}.$$

We prove that A is empty by contradiction. Assume $A \neq \emptyset$. Let a be a minimal element of A for $\prec_R$; such an element exists since $\prec_R$ is Noetherian.

Let b and c be terms such that $a \rightarrow_R^* b$ and $a \rightarrow_R^* c$.

If both reductions occur in a single rewriting step, then, by local confluence, there exists d such that $b \rightarrow_R^* d$ and $c \rightarrow_R^* d$: b, c are joinable, contradicting $a \in A$.

Therefore, at least one reduction has length greater than one. Without loss of generality, there exist b_1, c_1 such that $b \leftarrow_R^* b_1 \leftarrow_R a \rightarrow_R c_1 \rightarrow_R^* c$.

Since $a \rightarrow_R b_1$ and $a \rightarrow_R c_1$ are single steps, local confluence yields e with $b_1 \rightarrow_R^* e$ and $c_1 \rightarrow_R^* e$.

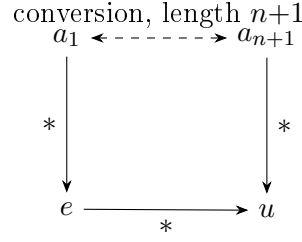
Because $c_1 \prec_R a$, minimality of a gives $c_1 \notin A$: c and e are joinable in some u . Similarly $b_1 \prec_R a$ gives $b_1 \notin A$: b and e are joinable in some f . Finally $e \prec_R a$ gives $e \notin A$: u and f are joinable in some g . It follows that b and c are joinable, contradicting $a \in A$.

Therefore A is empty: every terminating and locally confluent rewrite system is confluent. $\square$

Theorem (Confluence and termination). A terminating TRS is confluent if and only if it is locally confluent.

Proof. Newman's Lemma already establishes that a terminating and locally confluent TRS is confluent. The converse (a terminating confluent TRS is locally confluent) is immediate, since local confluence is a particular case of confluence. $\square$

Lemma (*confluence implies joinability of conversions*). If R is confluent, then for all terms a, b such that $a \leftrightarrow_R^* b$, there exists a term c such that $a \rightarrow_R^* c \leftarrow_R^* b$.



Intuitively, the induction shows that a zigzag conversion of any length can always be flattened into a single valley: the picture above illustrates the inductive step, replacing a conversion of length $n + 1$ by a common reduct u built from the shorter conversion's own common reduct e .

Proof. Let $P(n)$ be: "for every conversion sequence $a \leftrightarrow_R^* b$ of length n , there exists c with $a \rightarrow_R^* c \leftarrow_R^* b$."

Base case ($n = 0$). If $a \leftrightarrow_R^0 b$, then $a = b$; take $c = a$.

Induction step. Assume $P(n)$. Consider a conversion sequence of length $n + 1$, $a_1 \leftrightarrow_R a_2 \leftrightarrow_R \dots \leftrightarrow_R a_{n+1}$. By the induction hypothesis, there exists e with $a_1 \rightarrow_R^* e$ and $a_n \rightarrow_R^* e$. If $a_n \rightarrow_R a_{n+1}$, then by confluence e and a_{n+1} are joinable in some u , and $a_1 \rightarrow_R^* u$, $a_{n+1} \rightarrow_R^* u$. If $a_{n+1} \rightarrow_R a_n$, then $a_{n+1} \rightarrow_R a_n \rightarrow_R^* e$, so $a_{n+1} \rightarrow_R^* e$, together with $a_1 \rightarrow_R^* e$. In both cases $P(n + 1)$ holds. $\square$

Theorem (*Termination and Church–Rosser*). A terminating TRS R is Church–Rosser if and only if it is locally confluent (equivalently, confluent).

Proof. The previous lemma establishes Confluence $\Rightarrow$ Church–Rosser. We now prove the converse. Assume R satisfies Church–Rosser. Let a, b, c be such that $a \rightarrow_R^* b$ and $a \rightarrow_R^* c$. Then $c \leftarrow_R a \rightarrow_R^* b$, hence $c \leftrightarrow_R^* a \leftrightarrow_R^* b$, so $c \leftrightarrow_R^* b$. By Church–Rosser, there exists d with $c \rightarrow_R^* d$ and $b \rightarrow_R^* d$. Hence every pair of reductions from the same term is joinable, so R is confluent. $\square$

4 Unification

The Critical Pair Theorem, our next goal, will rely crucially on the existence of a *most general unifier* between two patterns — this is precisely how one detects, and resolves, the ambiguities that threaten local confluence. We therefore develop this notion now: we present the classical rule-based unification algorithm, prove its termination, and establish that it computes a most general unifier whenever one exists. This ensures that Section 5 rests on solid ground, rather than assuming unifiability as a black box.

Definition (*Unifier*). A substitution σ is a *unifier* of two terms s, t if $s\sigma = t\sigma$. A *most general unifier* (MGU) of s, t is a unifier σ such that for every unifier τ of s, t there exists a substitution θ with $\tau = \sigma \circ \theta$.

Definition (*Unification problem*). A *unification problem* is a finite set of equations $P = \{s_1 \doteq t_1, \dots, s_k \doteq t_k\}$ with $s_i, t_i \in \mathcal{T}(\Sigma, \mathcal{X})$. A substitution σ *solves* P if $s_i\sigma = t_i\sigma$ for every i . We write $\text{Sol}(P)$ for the set of unifiers of P .

Definition (*Unification algorithm*). The unification algorithm is a transition system on unification problems (together with an accumulated substitution, initially the identity), governed by the following rules, applied to an arbitrary equation $s \doteq t \in P$ chosen non-deterministically:

Delete. $\{s \doteq s\} \cup P \vdash P$.

Decompose. $\{f(s_1, \dots, s_n) \doteq f(t_1, \dots, t_n)\} \cup P \vdash \{s_1 \doteq t_1, \dots, s_n \doteq t_n\} \cup P$.

Clash. $\{f(s_1, \dots, s_n) \doteq g(t_1, \dots, t_m)\} \cup P \vdash \text{FAIL}$, if $f \neq g$.

Orient. $\{t \doteq x\} \cup P \vdash \{x \doteq t\} \cup P$, if $t \notin \mathcal{X}$ and $x \in \mathcal{X}$.

Eliminate. $\{x \doteq t\} \cup P \vdash \{x \doteq t\} \cup P[x := t]$, if $x \in \mathcal{X}$, $x \notin \text{Var}(t)$, and x occurs elsewhere in P (here $P[x := t]$ denotes P with every other occurrence of x replaced by t).

Occurs-check. $\{x \doteq t\} \cup P \vdash \text{FAIL}$, if $x \in \mathcal{X}$, $x \neq t$, and $x \in \text{Var}(t)$.

A problem P is in *solved form* if every equation is of the shape $x \doteq t$ with $x \in \mathcal{X}$ occurring exactly once in the whole of P (i.e. nowhere else, neither on a left- nor a right-hand side of another equation), in which case it directly defines a substitution $\sigma_P(x) := t$ (and $\sigma_P(y) := y$ for y not occurring on the left of an equation of P).

Lemma (*termination of the unification algorithm*). No infinite chain of transitions $P_0 \vdash P_1 \vdash P_2 \vdash \dots$ (other than immediately reaching FAIL or a solved form) exists.

Proof. Define a measure $\mu(P) = (n_1(P), n_2(P)) \in \mathbb{N} \times \mathbb{N}$, ordered lexicographically, where $n_1(P)$ is the number of distinct variables occurring in P that are *not* the left-hand side of a solved equation, and $n_2(P)$ is the total number of symbols occurring in P (summing the sizes of all s_i, t_i).

Delete strictly decreases n_2 (one equation, hence at least one symbol, is removed), and does not increase n_1 (no new variable is introduced, none is eliminated either, so n_1 is unchanged or decreases). Hence μ strictly decreases lexicographically.

Decompose replaces one equation between two terms of size ≥ 2 (at least the root symbol f plus the $n \geq 1$ arguments) by n equations between the immediate arguments: the total symbol count strictly decreases (the n occurrences of f -headed terms are replaced by their arguments, removing at least the two occurrences of f itself), while n_1 does not increase. Hence μ strictly decreases.

Orient leaves n_1, n_2 unchanged in magnitude but is applied at most once per equation before Decompose/Delete/Eliminate/Occurs-check apply (an equation $x \doteq t$ cannot be re-oriented, since Orient only fires on $t \doteq x$ with $t \notin \mathcal{X}$); we account for this by refining μ with a third, coarser component counting the number of equations not yet oriented correctly (of the form $t \doteq x$), which strictly decreases and is never increased by any other rule.

Eliminate strictly decreases n_1 : the variable x no longer occurs in P except as the (now unique) left-hand side of $x \doteq t$, since every other occurrence has been substituted away; no new variable is introduced (t 's variables were already present, as $P[x := t]$ only replaces occurrences of x , which does not introduce fresh variables). Hence μ strictly decreases lexicographically at the first component.

Clash and **Occurs-check** terminate the process immediately (FAIL).

Since μ takes values in a well-founded set ($\mathbb{N}^3$ with the lexicographic order, or $\mathbb{N}^2$ once Orient is accounted for separately as above) and strictly decreases at every non-terminal transition, no infinite chain of transitions exists. $\square$

Theorem (*Correctness of unification*). Let P_0 be a unification problem. Every maximal run of the unification algorithm from P_0 terminates either in FAIL, or in a solved form P defining a substitution σ_P . In the first case, $\text{Sol}(P_0) = \emptyset$. In the second, σ_P is a most general unifier of P_0 .

Proof. Termination of every run is Lemma [termination of the unification algorithm]. We show that every transition rule preserves the set of solutions, i.e. if $P \vdash P'$ (with $P' \neq \text{FAIL}$) then $\text{Sol}(P) = \text{Sol}(P')$, and that Clash/Occurs-check fire exactly when $\text{Sol}(P) = \emptyset$.

Delete. σ solves $\{s \doteq s\} \cup P$ iff $s\sigma = s\sigma$ (always true) and σ solves P : same solution set.

Decompose. σ solves $f(s_1, \dots, s_n) \doteq f(t_1, \dots, t_n)$ iff $f(s_1, \dots, s_n)\sigma = f(t_1, \dots, t_n)\sigma$ iff $s_i\sigma = t_i\sigma$ for every i (by injectivity of term formation: two terms with the same root and the same arguments, position by position, are equal, and conversely): same solution set as the decomposed system.

Clash. If $f \neq g$, no substitution can make $f(\dots)\sigma = g(\dots)\sigma$, since substitution never changes the root symbol of a non-variable term: $f(\dots)\sigma$ is rooted by f , $g(\dots)\sigma$ by g . Hence $\text{Sol}(P) = \emptyset$.

Orient. σ solves $t \doteq x$ iff $t\sigma = x\sigma$ iff $x\sigma = t\sigma$: same solution set as $x \doteq t$.

Eliminate. Suppose σ solves $\{x \doteq t\} \cup P$: then $x\sigma = t\sigma$, and for any equation $u \doteq v \in P$, $(u[x := t])\sigma = u\sigma[x := t\sigma\text{-substituted for } x\sigma] = u\sigma$ (since substituting x by t inside u and then applying σ is the same as applying σ directly, using $x\sigma = t\sigma$), so σ also solves $P[x := t]$, hence $\{x \doteq t\} \cup P[x := t]$. Conversely if σ solves $\{x \doteq t\} \cup P[x := t]$, then since $x\sigma = t\sigma$, for $u \doteq v \in P$, $u\sigma = (u[x := t])\sigma = v[x := t]\sigma \dots$ (symmetric argument) $= v\sigma$: σ solves $u \doteq v$. Hence σ solves P , so $\{x \doteq t\} \cup P$. Same solution set.

Occurs-check. If $x \in \text{Var}(t)$ and $x \neq t$, then t is (or contains as a proper subterm) a term strictly containing x ; any solution would require $x\sigma = t\sigma$, but $t\sigma$ has strictly more symbols than $x\sigma$ whenever x occurs properly inside t (an elementary induction on the structure of t , using that substitution only grows or preserves term size symbol by symbol), which is impossible for a fixed-point equation $x\sigma = t\sigma$ unless $x = t$. Hence $\text{Sol}(\{x \doteq t\} \cup P) = \emptyset$.

Since every transition preserves Sol (or correctly detects $\text{Sol} = \emptyset$), and $\text{Sol}(P_0) = \text{Sol}(P_{\text{final}})$ along any run, we conclude: if the run ends in FAIL, $\text{Sol}(P_0) = \emptyset$. If it ends in a solved form P , then $\text{Sol}(P_0) = \text{Sol}(P)$, and σ_P (reading off $\sigma_P(x) = t$ for each solved equation $x \doteq t$) is itself a solution of P (immediate verification, since each variable occurs exactly once as a left-hand side, so no conflicting requirement can arise), hence of P_0 : $\sigma_P \in \text{Sol}(P_0)$.

It remains to show σ_P is *most general*: let $\tau \in \text{Sol}(P_0) = \text{Sol}(P)$. For each solved equation $x \doteq t$ of P , $x\tau = t\tau$; define $\theta := \tau$ restricted appropriately (formally, since x occurs uniquely as a left-hand side, we may take $\theta = \tau$ itself and verify $\tau = \sigma_P \circ \tau$ on every variable: for x a left-hand side of P , $(\sigma_P \circ \tau)(x) = t\tau = x\tau$; for x not occurring as a left-hand side, $\sigma_P(x) = x$, so $(\sigma_P \circ \tau)(x) = x\tau$). Hence $\tau = \sigma_P \circ \tau$, witnessing that σ_P is a most general unifier. $\square$

Example. Unify $s = f(x, g(y))$ and $t = f(g(a), x)$ (with a a constant).

$$\begin{aligned}
\{f(x, g(y)) \doteq f(g(a), x)\} &\vdash_{\text{Decompose}} \{x \doteq g(a), g(y) \doteq x\} \\
&\vdash_{\text{Orient}} \{x \doteq g(a), x \doteq g(y)\} \\
&\vdash_{\text{Eliminate}} \{x \doteq g(a), g(a) \doteq g(y)\} \\
&\vdash_{\text{Decompose}} \{x \doteq g(a), a \doteq y\} \\
&\vdash_{\text{Orient}} \{x \doteq g(a), y \doteq a\}
\end{aligned}$$

This is a solved form, giving the most general unifier $\sigma(x) = g(a)$, $\sigma(y) = a$. One checks $s\sigma = f(g(a), g(a)) = t\sigma$.

5 Critical pairs

The property of confluence is very important for a TRS, especially for equational reasoning. It allows us to rewrite an expression in different ways and still obtain the same result. Generally, it is very difficult to obtain a confluent TRS directly. But in the special case of terminating TRSs, Newman's Lemma reduces the problem to ensuring local confluence. In this context, the objective becomes: how can we make a TRS locally confluent, in a way that is mechanically checkable?

Definition (Overlap). Let R be a term rewriting system. An *overlap* occurs when there exist rules $l_1 \rightarrow r_1$ and $l_2 \rightarrow r_2$ in R , a non-variable position $p \in \text{Pos}(l_2)$, and a substitution σ such that $l_2|_p\sigma = l_1\sigma$.

Definition (Critical pair). Let $l_1 \rightarrow_R r_1$ and $l_2 \rightarrow_R r_2$ be two rewrite rules sharing no variables (renamed apart if necessary). Suppose there exists a position $p \in \text{Pos}_{\mathcal{F}}(l_1)$ (i.e. p is not a variable position of l_1) and a most general unifier σ such that $l_1|_p\sigma = l_2\sigma$. This gives rise to two possible reductions of $l_1\sigma$: $l_1\sigma \rightarrow r_1\sigma$, and $l_1\sigma \rightarrow l_1\sigma[r_2\sigma]_p$. We call *critical pair* the pair $(r_1\sigma, l_1\sigma[r_2\sigma]_p)$. By convention, the trivial overlap of a rule with itself at the root position is excluded, as it always yields the trivial pair (r_1, r_1) .

Algorithm 1 ComputeCriticalPairs

Require: A term rewriting system $R = \{l_1 \rightarrow_R r_1, \dots, l_n \rightarrow_R r_n\}$

Ensure: The set $CP(R)$ of all (non-trivial) critical pairs of R

```
1: function COMPUTECRITICALPAIRS( $R$ )
2:    $CP \leftarrow \emptyset$ 
3:   for all rule  $(l_1 \rightarrow_R r_1) \in R$  do
4:     for all rule  $(l_2 \rightarrow_R r_2) \in R$  do
5:        $(l'_2, r'_2) \leftarrow \text{RENAME}(l_2 \rightarrow_R r_2)$ 
6:       for all position  $p \in \text{Pos}_{\mathcal{F}}(l_1)$  do
7:         if  $(l_1 \rightarrow_R r_1) = (l_2 \rightarrow_R r_2)$  and  $p = \varepsilon$  then
8:           continue
9:         end if
10:         $\sigma \leftarrow \text{UNIFY}(l_1|_p, l'_2)$ 
11:        if  $\sigma \neq \text{FAIL}$  then
12:           $CP \leftarrow CP \cup \{(\sigma(r_1), \sigma(l_1[r'_2]_p))\}$ 
13:        end if
14:      end for
15:    end for
16:  end for
17:  return  $CP$ 
18: end function
```

Note. Let (s, t) be a critical pair of R . It is *trivial* if $s \downarrow_R t$. In particular every critical pair with $s = t$ is trivial.

Definition (*Parallel rewriting*). Let R be a TRS. For a term t , a *parallel rewrite step* is a relation $t \Rightarrow_R t'$ defined as follows: there exists a finite non-overlapping set of positions $P = \{p_1, \dots, p_n\} \subseteq \text{Pos}(t)$, and for each i a rule $\ell_i \rightarrow r_i \in R$ and substitution σ_i with $t|_{p_i} = \ell_i \sigma_i$, such that

$$t' = t[r_1 \sigma_1]_{p_1} \cdots [r_n \sigma_n]_{p_n}.$$

Note. $\rightarrow_R \subseteq \Rightarrow_R \subseteq \rightarrow_R^*$.

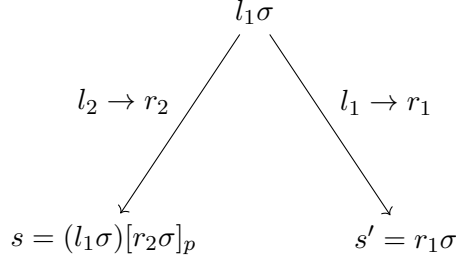
Definition (*context*). Let $\{\square_1, \dots, \square_n\}$ be fresh symbols, disjoint from $\Sigma \cup \mathcal{X}$, each treated as a constant. A *context* with holes $\square_1, \dots, \square_n$ is a term $C \in \mathcal{T}(\Sigma, \mathcal{X} \cup \{\square_1, \dots, \square_n\})$. Given terms $s_1, \dots, s_n$, $C[s_1, \dots, s_n]$ denotes the term obtained by simultaneously replacing every occurrence of $\square_k$ by s_k .

Nota. C is *linear* in $\square_1, \dots, \square_n$ if each hole occurs at exactly one position of C , denoted p_k .

Theorem (*Critical Pair Theorem (Knuth–Bendix)*). A terminating TRS is confluent if and only if both terms of each of its critical pairs reduce to a common term.

Proof. Let R be a terminating TRS.

($\Rightarrow$) Assume R is confluent. Let (s, s') be a critical pair, arising from rules $l_1 \rightarrow r_1, l_2 \rightarrow r_2$, position p , and most general unifier σ , with $s = (l_1 \sigma)[r_2 \sigma]_p$ and $s' = r_1 \sigma$:



Since $l_1\sigma \rightarrow_R s$ and $l_1\sigma \rightarrow_R s'$, confluence gives a term u with $s' \rightarrow_R^* u \leftarrow_R^* s$: every critical pair is joinable.

($\Leftarrow$) Assume every critical pair is joinable. By Newman's Lemma, it suffices to show local confluence: every local peak $b \leftarrow_R a \rightarrow_R c$ is joinable.

Let $A = \{a \mid \exists b, c : b \leftarrow_R a \rightarrow_R c, \nexists d, b \rightarrow_R^* d \leftarrow_R^* c\}$. Suppose $A \neq \emptyset$; since the strict subterm ordering is well-founded, fix $a \in A$ minimal for the strict subterm relation among the elements of A , and fix b, c as above via rules $l_1 \rightarrow r_1, l_2 \rightarrow r_2$, positions $p, q \in \text{Pos}(a)$, substitutions σ_1, σ_2 , with $a|_p = l_1\sigma_1$, $a|_q = l_2\sigma_2$, $b = a[r_1\sigma_1]_p$, $c = a[r_2\sigma_2]_q$.

We study the relative positions of p and q : we say $p \leq q$ if p is a prefix of q , i.e. $q = p \cdot q'$ (possibly $q' = \varepsilon$, giving $p = q$); and p, q are disjoint if neither is a prefix of the other. Exactly one of $p \leq q$, $q \leq p$, or p, q disjoint holds.

Case 1 (disjoint positions). Replacing a at p does not modify the subterm at q : $(a[r_1\sigma_1]_p)|_q = a|_q = l_2\sigma_2$, so the redex at q survives, giving $a[r_1\sigma_1]_p \rightarrow_R a[r_1\sigma_1]_p[r_2\sigma_2]_q$. Symmetrically, $a[r_2\sigma_2]_q \rightarrow_R a[r_2\sigma_2]_q[r_1\sigma_1]_p$. Since p, q are disjoint, the two replacements act on independent subtrees and commute: $a[r_1\sigma_1]_p[r_2\sigma_2]_q = a[r_2\sigma_2]_q[r_1\sigma_1]_p =: d$. Then $b \rightarrow_R d$ and $c \rightarrow_R d$: the peak is joinable.

Case 2 (p and q comparable). Without loss of generality, $p \leq q$ (the case $q \leq p$ is symmetric, exchanging the roles of the two rules). Write $q = p \cdot q'$ with $q' \in \text{Pos}(l_1\sigma_1)$.

Case 2.1 (q' not below a variable of l_1). Then $q' \in \text{Pos}(l_1)$ and $(l_1\sigma_1)|_{q'} = (l_1|_{q'})\sigma_1$.

If $q' = \varepsilon$: $l_1\sigma_1 = l_2\sigma_2$, the two rules apply to the same redex. If the two rules are identical up to a renaming of variables (bijective ρ with $l_2 = l_1\rho$, $r_2 = r_1\rho$), then $r_1\sigma_1 = r_2\sigma_2$, so $b = c$, trivially joinable. Otherwise, unifying l_1, l_2 yields a most general unifier μ and critical pair $(r_1\mu, r_2\mu)$; by hypothesis joinable, giving d with $r_1\mu \rightarrow_R^* d$, $r_2\mu \rightarrow_R^* d$. There exists θ with $\sigma_1 = \mu\theta$ on $\text{Var}(l_1)$, $\sigma_2 = \mu\theta$ on $\text{Var}(l_2)$; by stability under substitution and context, $b \rightarrow_R^* a[d\theta]_p$ and $c \rightarrow_R^* a[d\theta]_p$.

If $q' \neq \varepsilon$: q' is a non-variable position of l_1 , so unifying $l_1|_{q'}$ and l_2 yields a most general unifier μ and critical pair $(r_1\mu, l_1[r_2]_{q'}\mu)$; by hypothesis joinable in some d . As above, via θ , $b \rightarrow_R^* a[d\theta]_p$ and $c \rightarrow_R^* a[d\theta]_p$.

Case 2.2 (q' below a variable of l_1). There exist $x \in \text{Var}(l_1)$, $p_x \in \text{Pos}(l_1)$ with $l_1|_{p_x} = x$, and $q'' \in \text{Pos}(x\sigma_1)$ with $q' = p_x \cdot q''$, $(x\sigma_1)|_{q''} = l_2\sigma_2$. Define $\sigma'_1(x) = (x\sigma_1)[r_2\sigma_2]_{q''}$, $\sigma'_1(y) = \sigma_1(y)$ for $y \neq x$. Then $c = a[r_2\sigma_2]_q = a[l_1\sigma'_1]_p \rightarrow_R a[r_1\sigma'_1]_p =: w$. Since $x\sigma_1 \rightarrow_R x\sigma'_1$ is precisely the step $l_2\sigma_2 \rightarrow r_2\sigma_2$, rewriting each of the (possibly $k \geq 0$) occurrences of $x\sigma_1$ inside $r_1\sigma_1$ gives $r_1\sigma_1 \rightarrow_R^* r_1\sigma'_1$, hence, by stability under context, $b = a[r_1\sigma_1]_p \rightarrow_R^* a[r_1\sigma'_1]_p = w$: the peak is joinable.

All local peaks are joinable: R is locally confluent, and by Newman's Lemma, R is confluent. $\square$

6 Completion

Not every TRS is confluent, and not every equational theory admits an obviously confluent orientation. The *Knuth–Bendix completion procedure* attempts to construct, from an arbitrary equational presentation, a convergent TRS defining the same equational theory — by repeatedly detecting and resolving the failures of local confluence exhibited by critical pairs (Section 5), whose computation in turn rests on the unification algorithm of Section 4.

Definition (*Equivalent TRS*). Two TRS R, S are *equivalent* if $\rightarrow_R^* = \rightarrow_S^*$.

Definition (*Reduction order*). A *reduction order* on $\mathcal{T}(\Sigma, \mathcal{X})$ is a strict order $\succ$ that is (i) well-founded, (ii) stable under substitution ($s \succ t \Rightarrow s\sigma \succ t\sigma$), (iii) stable under context ($s \succ t \Rightarrow C[s] \succ C[t]$). A rule $l \rightarrow r$ is *compatible* with $\succ$ if $l \succ r$; a TRS R is compatible with $\succ$ if every rule is.

Lemma (*compatible TRS terminate*). If R is compatible with a reduction order $\succ$, then R terminates.

Proof. An infinite reduction $t_0 \rightarrow t_1 \rightarrow \dots$ would give, by stability under substitution and context at each redex, $t_0 \succ t_1 \succ \dots$, an infinite $\succ$ -decreasing sequence, contradicting well-foundedness. $\square$

Definition (*Configuration transition relation*). Fix a finite E_0 and a reduction order $\succ$. A *configuration* is a pair (E, R) , E a set of equations, R a TRS. We write $(E, R) \vdash (E', R')$ to mean (E', R') is obtained from (E, R) by one application of one of the six inference rules below, and $\vdash^*$ for its reflexive-transitive closure.

Definition (*Knuth–Bendix completion*). The completion procedure is the transition system on configurations (E, R) , starting from $(E_0, \emptyset)$, governed by the following rules.

Orient. $(E \cup \{s \approx t\}, R) \vdash (E, R \cup \{s \rightarrow t\})$ if $s \succ t$ (symmetrically for $t \succ s$).

Delete. $(E \cup \{s \approx s\}, R) \vdash (E, R)$.

Simplify. $(E \cup \{s \approx t\}, R) \vdash (E \cup \{s' \approx t\}, R)$ if $s \rightarrow_R s'$.

Compose. $(E, R \cup \{l \rightarrow r\}) \vdash (E, R \cup \{l \rightarrow r'\})$ if $r \rightarrow_R r'$.

Collapse. $(E, R \cup \{l \rightarrow r\}) \vdash (E \cup \{l' \approx r\}, R)$ if $l \rightarrow_R l'$ using another rule of R at a non-root, non-variable position of l .

Deduce. $(E, R) \vdash (E \cup \{s \approx t\}, R)$ if (s, t) is a critical pair of R .

A run $(E_0, \emptyset) \vdash (E_1, R_1) \vdash \dots$ *succeeds* if it reaches $(\emptyset, R)$: R is then the *completion* of E_0 w.r.t. $\succ$.

Definition (*equational closure of a configuration*). For (E, R) , let $=_{E,R}$ be the smallest congruence, stable under substitution, containing $\rightarrow_R$ and every instance $s\sigma \approx t\sigma$ for $s \approx t \in E$.

Lemma (*soundness of completion steps*). If $(E, R) \vdash (E', R')$, then $=_{E,R} =_{E',R'}$.

Proof. *Orient:* the generator $s\sigma \approx t\sigma$ is reclassified as $s\sigma \rightarrow t\sigma$; unchanged as a relation. *Delete:* $(s\sigma, s\sigma)$ is already given by reflexivity. *Simplify:* since $s \rightarrow_R s'$ holds in both

configurations, $s\sigma \approx t\sigma$ and $s'\sigma \approx t\sigma$ are interderivable via $s\sigma \rightarrow_R s'\sigma \approx t\sigma$. *Compose*: similarly, using the shared reduction $r\sigma \rightarrow_R^* r'\sigma$. *Collapse*: using the shared $l\sigma \rightarrow_R^* l'\sigma \approx r\sigma$. *Deduce*: by construction, a critical pair (s, t) satisfies $s \leftarrow_R u \rightarrow_R t$ for a common ancestor u , so $s \approx t$ is already derivable from R alone. $\square$

Theorem (*Correctness of Knuth–Bendix completion*). If a run $(E_0, \emptyset) \vdash^* (\emptyset, R)$ succeeds, then R is convergent and equivalent to E_0 : $\rightarrow_R^* =_{E_0}$.

Proof. Termination. Every rule of the final R was introduced by Orient (compatible with $\succ$); Compose only replaces a right-hand side by a $\succ$ -smaller one; Collapse only removes rules. So R is compatible with $\succ$, hence terminates.

Confluence. By soundness applied along the whole run, $=_{E_0, \emptyset} =_{\emptyset, R}$. By Deduce, every critical pair of the final R was recorded as a pending equation at some point; since the run ends with $E = \emptyset$, every such pair (s, t) satisfies $s \leftrightarrow_R^* t$. Since R terminates, Newman’s Lemma converts this convertibility into joinability once local confluence is known; by the Critical Pair Theorem, joinability of every critical pair gives local confluence of R , hence, with termination, confluence by Newman’s Lemma.

Equivalence. By soundness, $=_{E_0} =_{E_0, \emptyset} =_{\emptyset, R} \leftrightarrow_R^*$. Since R is confluent, $\rightarrow_R^* =_{E_0}$. $\square$

Remark. Limits of the procedure. A run of completion need not succeed: at any Deduce step the resulting equation may fail to be orientable by the fixed order $\succ$ (a *failure* of completion, requiring a different or more powerful order), or the procedure may produce infinitely many rules without ever exhausting E (*divergence*, distinct from non-termination of the TRS produced). This is not a defect of our presentation: it reflects the genuine limits of the method, and is one of the motivations for variants such as unfailing completion, which we do not develop here.

7 Modularity

We now study the extent to which the properties introduced above are preserved under *disjoint unions* of term rewriting systems: given two systems $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ with $\Sigma_1 \cap \Sigma_2 = \emptyset$, each individually enjoying a property P , does $R_1 \cup R_2$ also enjoy P ? We show that the answer is a resounding *yes* for confluence (the theorem on the union of confluent TRS), and an equally resounding *no* for termination in general (the proposition on non-modularity of termination) — unless left-linearity is additionally assumed (the theorem on the union of convergent left-linear TRS). We also treat, as a structurally related but independent question, the confluence of *orthogonal* systems (Section 7.5).

Theorem (*Union of confluent TRS*). The union of two confluent TRS is confluent if they don’t share any function symbol (or constant).

Note. The advantage of having distinguished signatures is that in both systems no rule of one system rewrites a term into the other system’s territory in an uncontrolled way. On the other hand, the union means that we can have *mixed* terms, combining symbols of both signatures. The whole difficulty of this theorem lies in analysing how rewriting acts on such mixed terms.

7.1 Aliens, skeletons, and rank

Definition (*pure term*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$. A term t is *pure* for R_i if all of its symbols belong to Σ_i : $t \in T(\Sigma_i, \mathcal{X})$.

Definition (*Alien term*). Let $t \in \mathcal{T}(\Sigma, \mathcal{X})$ be a term whose root belongs to Σ_i (we say that t is *i-rooted*: $\text{root}(t) \in \Sigma_i$). A subterm $s = t|_p$ (for some $p \in \text{Pos}(t)$, $p \neq \varepsilon$) is called an *alien* of t (for R_i) if $\text{root}(s) \in \Sigma_j$, $j \neq i$, while every proper prefix of p carries a symbol of Σ_i . Equivalently: s is a maximal subterm of t whose root does not belong to Σ_i . No condition is imposed on the symbols occurring strictly below the root of s : an alien may itself contain further aliens (rooted back in Σ_i , or beyond).

Example. Let $\Sigma_1 = \{f, a, b\}$, $\Sigma_2 = \{g\}$, and $t = f(g(f(a), y), b)$. Then t is 1-rooted, and $g(f(a), y)$ is an alien of t (for R_1) — even though it itself contains $f(a)$, which is rooted back in Σ_1 . This illustrates why no purity condition is imposed on the interior of an alien.

Definition (*term preserver*). Let $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ be two rewrite systems with disjoint signatures. A rewrite rule $\ell \rightarrow r \in \mathcal{R}_1$ is a *term preserver* if $\mathcal{F}(r) \subseteq \Sigma_1 \cup \text{Var}(\ell)$ (it introduces no new symbol from Σ_2). Concretely, for a step $t \xrightarrow[p]{\ell \rightarrow r} t'$: any alien of t outside p is unchanged in t' , and any alien inside $t|_p$ is only substituted for a variable of ℓ , hence reappears unmodified (possibly copied or deleted, never altered). R_1 may rearrange, duplicate, or erase aliens from R_2 , but never fabricate a new one.

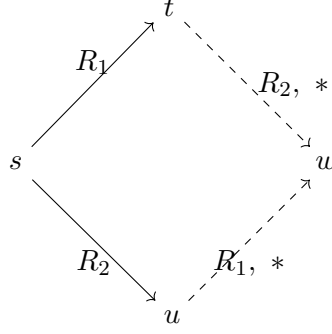
Definition (*collapsing rule*). A rule $\ell \rightarrow_R r \in \mathcal{R}_i$ is *collapsing* if $r \in \mathcal{X}$. A rewrite step is *collapsing* if it uses a collapsing rule.

Nota. A term t is *preserved* (for $R_1 \cup R_2$) if no reduction sequence issued from t , finite or infinite, contains a collapsing step.

Definition (*skeleton decomposition*). Let t be i -rooted with aliens $s_1, \dots, s_n$ at (pair-wise disjoint) positions $p_1, \dots, p_n$. There exists a unique linear context $C \in \mathcal{T}(\Sigma_i, \mathcal{X} \cup \{\square_1, \dots, \square_n\})$ with holes at exactly $p_1, \dots, p_n$ such that $t = C[s_1, \dots, s_n]$. We call C the *skeleton* of t .

7.2 The mixed local peak

Lemma (1). Let $R_1 = (\Sigma_1, \mathcal{R}_1)$ and $R_2 = (\Sigma_2, \mathcal{R}_2)$ be two TRS with $\Sigma_1 \cap \Sigma_2 = \emptyset$. Then $R_1 \cup R_2$ satisfies: for every local peak $t \xrightarrow{R_1} s \rightarrow_{R_2} u$, there exists w such that $t \xrightarrow{*}_{R_2} w$ and $u \xrightarrow{*}_{R_1} w$.



Proof. By definition of a rewrite step, there exist rules $l_1 \rightarrow_R r_1 \in \mathcal{R}_1$, $l_2 \rightarrow_R r_2 \in \mathcal{R}_2$, positions $p, q \in \text{Pos}(s)$, substitutions σ_1, σ_2 with $s|_p = l_1\sigma_1$, $s|_q = l_2\sigma_2$, $t = s[r_1\sigma_1]_p$, $u = s[r_2\sigma_2]_q$.

Preliminary claim. Let $i \in \{1, 2\}$, $l \rightarrow_R r \in \mathcal{R}_i$, σ a substitution, $q' \in \text{Pos}(l\sigma)$. Exactly one of the following holds: (a) q' is not below a variable of l , in which case $q' \in \text{Pos}(l)$ and $\text{root}((l\sigma)|_{q'}) = \text{root}(l|_{q'}) \in \Sigma_i$; (b) q' is below a variable of l : there exist $x \in \text{Var}(l)$, $p_x \in \text{Pos}(l)$ with $l|_{p_x} = x$, and $q'' \in \text{Pos}(x\sigma)$, such that $q' = p_x \cdot q''$, $(l\sigma)|_{q'} = (x\sigma)|_{q''}$. This follows by a straightforward induction on q' , using that $l \notin \mathcal{X}$ and $l \in T(\Sigma_i, \mathcal{X})$.

The case $p = q$ cannot occur. If $p = q$, then $s|_p = l_1\sigma_1 = l_2\sigma_2$. Applying the preliminary claim to each side at $q' = \varepsilon$ (always case (a), since ε has no proper prefix) gives $\text{root}(l_1\sigma_1) = \text{root}(l_1) \in \Sigma_1$ and $\text{root}(l_2\sigma_2) = \text{root}(l_2) \in \Sigma_2$, contradicting $\Sigma_1 \cap \Sigma_2 = \emptyset$.

Case $p \parallel q$ (disjoint). Replacing at p leaves the subterm at q unchanged: $(s[r_1\sigma_1]_p)|_q = s|_q = l_2\sigma_2$, so $s[r_1\sigma_1]_p \rightarrow_{R_2} s[r_1\sigma_1]_p[r_2\sigma_2]_q$. Symmetrically $s[r_2\sigma_2]_q \rightarrow_{R_1} s[r_2\sigma_2]_q[r_1\sigma_1]_p$. The two replacements act on disjoint subtrees, hence commute: let w be the common term; $t \rightarrow_{R_2} w$, $u \rightarrow_{R_1} w$.

Case $p < q$. Write $q = p \cdot q'$, $q' \in \text{Pos}(l_1\sigma_1)$. By the preliminary claim, case (a) would give $\text{root}((l_1\sigma_1)|_{q'}) \in \Sigma_1$, contradicting $(l_1\sigma_1)|_{q'} = s|_q = l_2\sigma_2$ (rooted in Σ_2). So case (b) holds: $q' = p_x \cdot q''$, $l_1|_{p_x} = x$, $(x\sigma_1)|_{q''} = l_2\sigma_2$. Define $\sigma'_1(x) = (x\sigma_1)[r_2\sigma_2]_{q''}$, $\sigma'_1(y) = \sigma_1(y)$ for $y \neq x$. Then $u = s[r_2\sigma_2]_q = s[l_1\sigma'_1]_p \rightarrow_{R_1} s[r_1\sigma'_1]_p =: w$, and since $x\sigma_1 \rightarrow_{R_2} x\sigma'_1$ is exactly the step $l_2\sigma_2 \rightarrow r_2\sigma_2$, rewriting each occurrence of $x\sigma_1$ inside $r_1\sigma_1$ gives $t = s[r_1\sigma_1]_p \rightarrow_{R_2}^* s[r_1\sigma'_1]_p = w$.

Case $q < p$ is symmetric to the previous case, exchanging the roles of the two rules.

Since the cases exhaust all possibilities, every local peak $t \leftarrow_{R_1} s \rightarrow_{R_2} u$ is joinable. $\square$

Lemma (*extension of confluence to extra variables*). Let $R_i = (\Sigma_i, \mathcal{R}_i)$ be confluent on $\mathcal{T}(\Sigma_i, \mathcal{X})$, and let $\mathcal{X}' \supseteq \mathcal{X}$. Then R_i is confluent on $\mathcal{T}(\Sigma_i, \mathcal{X}')$.

Proof. Let $C \rightarrow_{R_i}^* C_1$, $C \rightarrow_{R_i}^* C_2$ in $\mathcal{T}(\Sigma_i, \mathcal{X}')$. Let $\square_1, \dots, \square_n$ be the variables of $\mathcal{X}' \setminus \mathcal{X}$ occurring in C . Since every rule's left-hand side is non-variable, no rule matches a redex rooted at a $\square_k$; rewriting propagates the $\square_k$'s exactly as ordinary variables of $\mathcal{X}$. Renaming $\square_1, \dots, \square_n$ to fresh variables of $\mathcal{X}$ via a bijection ρ yields $C^\rho \rightarrow_{R_i}^* C_1^\rho$, $C^\rho \rightarrow_{R_i}^* C_2^\rho$ in $\mathcal{T}(\Sigma_i, \mathcal{X})$. By confluence there, D with $C_1^\rho \rightarrow_{R_i}^* D$, $C_2^\rho \rightarrow_{R_i}^* D$; applying ρ^{-1} , $C_1 \rightarrow_{R_i}^* D\rho^{-1}$, $C_2 \rightarrow_{R_i}^* D\rho^{-1}$. $\square$

Lemma (*skeleton-alien commutation*). Let $t = C[s_1, \dots, s_n]$ be i -rooted, and let $t \xrightarrow[p]{l \rightarrow r} t'$

be a single step. Exactly one holds: (a) p occurs strictly below some p_k : then $l \rightarrow r \in \mathcal{R}_j$ (j the index of s_k), $s_k \rightarrow_{R_j} s'_k$, $t' = C[s_1, \dots, s'_k, \dots, s_n]$, and this step is non-collapsing whenever t is preserved. (b) p does not occur below any p_k : then $l \rightarrow r \in \mathcal{R}_i$, and $C \rightarrow_{R_i} C'$ (a skeleton step, $\square_k$'s treated as variables), $t' = C'[s_1, \dots, s_n]$, unless $p = p_k$ and $l \rightarrow r$ is collapsing, which cannot occur when t is preserved.

Proof. Since $p_1, \dots, p_n$ are pairwise disjoint and p is a single position, either p is strictly below some unique p_k , or $p = p_k$, or p is disjoint from every p_k , or p is a strict prefix of some p_k (in which case $t|_p$ is i -rooted by maximality of the alien positions, falling under (b)). This exhausts all cases.

Case (a). $t|_p = s_k|_{p'}$ for $p = p_k \cdot p'$, so $\text{root}(t|_p) \in \Sigma_j$ and $l \rightarrow r \in \mathcal{R}_j$. Then $t' = C[s_1, \dots, s_k[r\sigma]_{p'}, \dots, s_n]$. If t is preserved, every step of every reduction from t is non-collapsing, in particular this one.

Case (b). $t|_p$ is matched entirely within C (the $\square_k$'s acting as opaque constants, since σ only instantiates variables of $\mathcal{X}$); the step is exactly a step of C viewed over the extended variable set. If $p = p_k$ and $l \rightarrow r$ is collapsing, $t' = t[\sigma(x)]_{p_k}$ exposes (part of) s_k at the root of the replaced subtree: this is a collapsing step, excluded when t is preserved. $\square$

Lemma (*purity is preserved under own-system rewriting*). Let $s \in \mathcal{T}(\Sigma_j, \mathcal{X})$ be pure for R_j , and $s \rightarrow_{R_j} s'$. Then $s' \in \mathcal{T}(\Sigma_j, \mathcal{X})$.

Proof. Write $s \xrightarrow[q]{\ell \rightarrow r} s'$ with $\ell, r \in \mathcal{T}(\Sigma_j, \mathcal{X})$. Every symbol of r is in Σ_j ; every subterm of s substituted for a variable of ℓ is itself pure (a subterm of a pure term is pure). Hence $r\sigma$, and thus s' , is pure. $\square$

Lemma (*subterms of preserved terms are preserved*). Let t be preserved and $s \sqsubset t$. Then s is preserved.

Proof. Suppose a reduction $s = s_0 \rightarrow s_1 \rightarrow \dots$ contains a collapsing step $s_k \rightarrow s_{k+1}$. Writing $t = C[s]$, stability under context reinjects each step as $C[s_l] \rightarrow C[s_{l+1}]$, using the same rule at the same relative position, hence the same collapsing status, producing a reduction from t containing a collapsing step: contradiction. $\square$

7.3 Rank and the induction measure

Definition (*strict subterm order*). We write $s \sqsubset t$ if there exists $p \in \text{Pos}(t)$, $p \neq \varepsilon$, with $s = t|_p$; and $s \sqsubseteq t$ if $s \sqsubset t$ or $s = t$.

Lemma (*well-foundedness of the strict subterm order*). $\sqsubset$ is well-founded on $\mathcal{T}(\Sigma, \mathcal{X})$.

Proof. Let $|t|$ count the total number of symbols of t (defined by structural recursion, which terminates since every term is built in finitely many steps). If $s \sqsubset t$ via a position of length 1, an immediate computation gives $|s| < |t|$; by induction on the length of the position, $|s| < |t|$ for every $s \sqsubset t$. An infinite $\sqsubset$ -decreasing chain would give an infinite strictly decreasing sequence of positive integers, impossible. $\square$

Definition (*rank*). We define $\text{rank}(t) \in \mathbb{N}^*$ by well-founded recursion on $\sqsubset$:

$$\text{rank}(t) = \begin{cases} 1 & \text{if } t \text{ is pure (for } R_1 \text{ or } R_2), \\ 1 + \max\{\text{rank}(s) : s \text{ alien of } t\} & \text{otherwise.} \end{cases}$$

Lemma (*strict rank decrease for aliens*). If s is an alien of t , then $\text{rank}(s) < \text{rank}(t)$.

Proof. Since t is not pure (it has an alien), $\text{rank}(t) = 1 + \max\{\text{rank}(s') : s' \text{ alien of } t\} \geq 1 + \text{rank}(s) > \text{rank}(s)$. $\square$

Example. Let $\Sigma_1 = \{f\}$, $\Sigma_2 = \{g\}$, $t = f(g(f(a), y), b)$ as before. The alien of t is $s_1 = g(f(a), y)$, which is not pure (it contains $f(a)$, rooted in Σ_1): its own alien is $f(a)$, which is pure. Hence $\text{rank}(f(a)) = 1$, $\text{rank}(g(f(a), y)) = 2$, $\text{rank}(t) = 3$. This illustrates why an unbounded number of alternations between the two signatures is possible, and why rank (not merely "pure vs. not pure") is the correct induction measure.

Lemma (*reduction decomposition*). Let $t = C_0[s_1, \dots, s_n]$ be preserved, i -rooted. Let $t = t_0 \rightarrow t_1 \rightarrow \dots \rightarrow t_m$ be any finite reduction. Then there exist, for each k , a skeleton C_k and a list of aliens $(u_1^{(k)}, \dots)$ with $t_k = C_k[u_1^{(k)}, \dots]$, and for each $k < m$ exactly one of: (a) $C_{k+1} = C_k$ and a unique alien is rewritten internally; (b) $C_k \rightarrow_{R_i} C_{k+1}$ (a skeleton step), with the alien list updated by duplication/erasure. Consequently, discarding type-(a) steps yields $C_0 \rightarrow_{R_i}^* C_m$.

Proof. Induction on m , applying Lemma [skeleton-alien commutation] at each step: the exceptional collapsing sub-case of (b) cannot occur since t is preserved, so every step of the reduction falls under case (a) or (b) as stated. $\square$

Lemma (*Confluence of preserved terms*). Let R_1, R_2 be confluent TRS with $\Sigma_1 \cap \Sigma_2 = \emptyset$. Every preserved term t is confluent for $R_1 \cup R_2$.

Proof. Let $R = R_1 \cup R_2$ and $T = \{t \text{ preserved} \mid \exists t_1, t_2 : t_1 \xrightarrow{+}_R t \xrightarrow{+}_R t_2, \nexists w, t_1 \rightarrow_R^* w \xleftarrow{*}_R t_2\}$. Suppose $T \neq \emptyset$; let $n_0 = \min\{\text{rank}(t) : t \in T\}$, fix $t \in T$ with $\text{rank}(t) = n_0$ and witnesses t_1, t_2 .

Case $n_0 = 1$. t is pure, say for R_1 . No rule of R_2 matches any position of t , so every R -step from t is an R_1 -step, and by Lemma [purity is preserved under own-system rewriting], every term reachable from t stays pure. Confluence of R_1 gives w joining t_1, t_2 : contradiction.

Case $n_0 > 1$. $t = C_0[s_1, \dots, s_n]$ ($n \geq 1$). By Lemma [strict rank decrease for aliens] and Lemma [subterms of preserved terms are preserved], each s_k has $\text{rank} < n_0$ and is preserved; by minimality of n_0 , each s_k is confluent for R .

Apply Lemma [reduction decomposition] to $t \rightarrow_R^* t_1$ and $t \rightarrow_R^* t_2$: $C_0 \rightarrow_{R_1}^* C^{(1)}$, $t_1 = C^{(1)}[u^{(1)}]$; $C_0 \rightarrow_{R_1}^* C^{(2)}$, $t_2 = C^{(2)}[u^{(2)}]$. By Lemma [extension of confluence to extra variables], R_1 is confluent with the holes as extra variables: C^* with $C^{(1)} \rightarrow_{R_1}^* C^*$, $C^{(2)} \rightarrow_{R_1}^* C^*$. By stability under substitution, $t_1 \rightarrow_R^* C^*[u^{(1)}]$, $t_2 \rightarrow_R^* C^*[u^{(2)}]$.

Each hole h of C^* traces to a unique alien origin s_k ; the two fillings $u_h^{(1)}, u_h^{(2)}$ are reachable from s_k , which is confluent: w_h joins them. Setting $w := C^*[(w_h)_h]$, $t_1 \rightarrow_R^* w$ and $t_2 \rightarrow_R^* w$: contradiction.

Both cases yield a contradiction, so $T = \emptyset$. $\square$

Lemma (*rank is non-increasing under rewriting*). If $t \rightarrow_{R_1 \cup R_2} t'$, then $\text{rank}(t') \leq \text{rank}(t)$.

Proof. Write $t \xrightarrow[q]{\ell \rightarrow r} t'$ with $\ell, r \in \mathcal{T}(\Sigma_i, \mathcal{X})$. The right-hand side $r\sigma$ is built from r (purely Σ_i) and copies of subterms already occurring in $t|_q$: no alien of t' exhibits an alternation pattern deeper than what was already present in t . A structural induction on the construction of rank then gives $\text{rank}(t') \leq \text{rank}(t)$. $\square$

Lemma (*first-switch decomposition*). Let t be i -rooted with skeleton $t = C_0[s_1, \dots, s_n]$, and $t \rightarrow^* t'$ any finite reduction. Exactly one holds: (N) no collapsing step exposes the currently active root, and $t' = C'[u_1, \dots, u_k]$ with $C_0 \rightarrow_{R_i}^* C'$; (S) a type-(N) prefix $t \rightarrow^* v = C'[u_1, \dots, u_k]$, followed by a single collapsing step exposing a filling u_{k_0} , followed by $u_{k_0} \rightarrow^* t'$, with $\text{rank}(u_{k_0}) < \text{rank}(t)$ and $\text{rank}(t') \leq \text{rank}(u_{k_0})$.

Proof. Repeated application of Lemma [skeleton-alien commutation]: as long as no step collapses the active root, every step is of type (N). The first collapsing step (if any) at the active root gives case (S); by Lemma [strict rank decrease for aliens] the rank strictly drops, and by Lemma [rank is non-increasing under rewriting] it never climbs back, so no second "root-collapse relative to C_0 " needs distinguishing. $\square$

7.4 Proof of the modularity of confluence

Proof of the theorem on the union of confluent TRS. By well-founded induction on $\text{rank}(t)$: we show, for every n_0 , that every term of rank n_0 is confluent for $R := R_1 \cup R_2$, assuming (IH) that every term of rank $< n_0$ is confluent for R .

Case $n_0 = 1$. As in the lemma on confluence of preserved terms, Case $n_0 = 1$: confluence of t reduces directly to confluence of R_1 .

Case $n_0 > 1$. Let $t = C_0[s_1, \dots, s_n]$, and consider a peak $t_1 \xrightarrow{*} t \xrightarrow{*} t_2$. By Lemma [first-switch decomposition], each branch is of type (N) or (S).

(N,N). Exactly the argument of the lemma on confluence of preserved terms, Case $n_0 > 1$, with (IH) replacing "preserved $\Rightarrow$ confluent" (each alien s_k has rank $< n_0$, hence is confluent for R by IH, with no need for the preservation hypothesis). This gives w joining t_1, t_2 .

(N,S) (the case (S,N) is symmetric). Write $t_1 = C^{(1)}[u^{(1)}]$ (type N); branch 2: $t \rightarrow^* v = C^{(2)}[u^{(2)}]$ (type N), then a collapsing step exposing $u_{k_0}^{(2)}$, then $u_{k_0}^{(2)} \rightarrow^* t_2$, with $\text{rank}(u_{k_0}^{(2)}) < n_0$.

Join t_1 and v exactly as in case (N,N): $y = C^*[y_1, \dots, y_n]$ with $t_1 \rightarrow_R^* y$, $v \rightarrow_R^* y$. Since the skeleton reduction $C^{(2)} \rightarrow_{R_i}^* C^*$ never touches hole contents, y_{k_0} is the reduct obtained by joining $u_{k_0}^{(2)}$ with its counterpart on branch 1. Since $\text{rank}(u_{k_0}^{(2)}) < n_0$, (IH) gives its confluence: applied to $t_2 \xrightarrow{*} u_{k_0}^{(2)} \rightarrow^* y_{k_0}$, this yields z with $t_2 \rightarrow_R^* z$, $y_{k_0} \rightarrow_R^* z$. Setting $w := C^*[y_1, \dots, z, \dots, y_n]$, $t_1 \rightarrow_R^* y \rightarrow_R^* w$ and $t_2 \rightarrow_R^* w$.

(S,S). Same construction, treating both collapses symmetrically via IH on each side (both ranks strictly below n_0).

In every case, t_1, t_2 are joinable: t is confluent for R . By well-founded induction, every term is confluent: $R_1 \cup R_2$ is confluent. $\square$

Nota (modularity). A property P of TRS is *modular* if, whenever R_1, R_2 both have P and $\Sigma_1 \cap \Sigma_2 = \emptyset$, $R_1 \cup R_2$ also has P . The theorem on the union of confluent TRS states precisely that confluence is modular. This modularity is far from automatic: the most instructive counterexample is termination, which is *not* modular, as we now show.

Proposition (*Terminating modularity*). Termination is not a modular property.

Proof. We exhibit two disjoint terminating TRS whose union does not terminate (Toyama, 1987). Let $\Sigma_1 = \{F, 0, 1\}$ (F ternary), $\Sigma_2 = \{G\}$ (G binary), and

$$\mathcal{R}_1 = \{ F(0, 1, x) \rightarrow F(x, x, x) \}, \quad \mathcal{R}_2 = \{ G(x, y) \rightarrow x, \quad G(x, y) \rightarrow y \}.$$

R_2 terminates. Each step replaces the redex $G(x, y)\sigma$ by $x\sigma$ or $y\sigma$, a strict subterm: $|s'| < |s|$ strictly, so no infinite reduction can exist.

R_1 terminates. A naive size argument fails, since $F(0, 1, x) \rightarrow F(x, x, x)$ can increase size. The correct argument (Toyama, 1987; Drosten, 1989) rests on the fact that $\mathcal{R}_1$ never freshly creates 0 or 1: every occurrence of the pattern $F(0, 1, \cdot)$ appearing after a step was already present, in duplicated form, before it. A minimal-counterexample argument on the depth of the first redex used along a hypothetical infinite reduction then yields a contradiction; we admit this without full detail, as it is not the subject of the present article.

The union does not terminate. Let $t_0 := F(G(0, 1), G(0, 1), G(0, 1))$. Using $G(x, y) \rightarrow x$ on the first occurrence: $t_0 \rightarrow F(0, G(0, 1), G(0, 1)) =: t_1$. Using $G(x, y) \rightarrow y$ on the (now second) occurrence: $t_1 \rightarrow F(0, 1, G(0, 1)) =: t_2$. Since t_2 matches $F(0, 1, x) \rightarrow F(x, x, x)$ with $\sigma(x) = G(0, 1)$: $t_2 \rightarrow F(G(0, 1), G(0, 1), G(0, 1)) = t_0$. Hence $t_0 \rightarrow t_1 \rightarrow t_2 \rightarrow t_0 \rightarrow \dots$ is a cyclic, infinite reduction.

Both R_1, R_2 terminate individually with $\Sigma_1 \cap \Sigma_2 = \emptyset$, yet the union admits an infinite reduction. Termination is not modular. $\square$

Remark. The mechanism at play is exactly the one isolated by Definition [collapsing rule]: both rules of $\mathcal{R}_2$ collapse a term to one of its own arguments, allowing an alien to be repeatedly re-exposed and duplicated across the two signatures without ever terminating. This is precisely why the proof of the theorem on the union of confluent TRS tracks collapsing steps so carefully, and why it could not simply invoke Newman's Lemma: $R_1 \cup R_2$ need not terminate even when R_1, R_2 both do. The argument instead proceeds by well-founded induction on rank, a measure entirely internal to the static structure of a term, insensitive to whether the rewriting relation terminates.

Theorem (*Union of convergent left-linear TRS*). Let R_1, R_2 be convergent TRS with $\Sigma_1 \cap \Sigma_2 = \emptyset$, such that every left-hand side of every rule of $R_1 \cup R_2$ is linear. Then $R_1 \cup R_2$ is convergent.

Proof. Confluence. Immediate from the theorem on the union of confluent TRS, applied to R_1, R_2 (confluent, since convergent). Left-linearity plays no role in this part.

Termination. This is a substantially harder result, due to Toyama, Klop, and Barendregt (*Termination for the direct sum of left-linear complete term rewriting systems*, J. ACM 42(6),

1995). Its proof requires an apparatus considerably beyond what we have developed here: a notion of *closed* term, a well-founded order on multisets of positions, and a delicate joint analysis of collapsing and duplicating steps — precisely the combination that the proposition on non-modularity of termination shows can defeat termination in the absence of left-linearity. Left-linearity is exactly the hypothesis that rules out the kind of inconsistent, non-confluent collapsing exhibited by $\mathcal{R}_2 = \{G(x, y) \rightarrow x, G(x, y) \rightarrow y\}$ in that counterexample (note that this $\mathcal{R}_2$ is itself left-linear, but is *not confluent*: $G(a, b)$ has two distinct normal forms a, b ; convergence, not merely left-linearity, is what is additionally required here). We do not reproduce the full argument, and instead refer the reader to the cited reference for the complete proof. $\square$

Remark. We flag this proof as *partial by design*: reproducing the Toyama–Klop–Barendregt argument in full would require substantially more space and machinery than the rest of this survey, and risks introducing subtle errors if compressed. We have chosen honesty over the appearance of completeness.

7.5 Orthogonal systems

We now turn to a second, independent route to confluence, applicable even to *non-terminating* systems, and requiring no disjointness of signatures: orthogonality. The guiding intuition is simple: if rewrite rules never overlap with one another (so a term can never be ambiguous as to which rule applies where) and never duplicate a variable on the left of a rule (so matching is unambiguous), then rewriting can be made confluent by a soft argument: any two ways of contracting a set of non-overlapping redexes in parallel can always be completed into contracting *every* redex of the term at once, and this "complete development" serves as a universal joining point.

Definition (*Orthogonal system*). A TRS R is *orthogonal* if it satisfies:

1. (*non-overlapping*) For any two rules $l_1 \rightarrow r_1, l_2 \rightarrow r_2 \in R$, there do not exist a non-variable position $p \in \text{Pos}(l_2)$ and a substitution σ with $l_2|_p = l_1\sigma$.
2. (*left-linearity*) Every left-hand side is linear (no variable occurs more than once).

Lemma (*unique redex identification under orthogonality*). Let R be orthogonal and $t|_p$ a redex. Then the rule $l \rightarrow r$ and substitution σ with $t|_p = l\sigma$ are unique.

Proof. If $t|_p = l_1\sigma_1 = l_2\sigma_2$, then taking $p' = \varepsilon$ in $\text{Pos}(l_2)$ gives an overlap of $l_1 \rightarrow r_1$ into $l_2 \rightarrow r_2$ at the root, forbidden unless the two rules coincide (the trivial self-overlap is not excluded by non-overlapping). Uniqueness of σ then follows from left-linearity: each variable of l occurs at a single position, so $\sigma(x) = t|_{p \cdot p_x}$ is forced. $\square$

Lemma ($\Rightarrow_R^*$ equals $\rightarrow_R^*$). $\Rightarrow_R^* = \rightarrow_R^*$.

Proof. $\rightarrow_R \subseteq \Rightarrow_R \subseteq \rightarrow_R^*$ (Definition [Parallel rewriting]); taking reflexive-transitive closures gives $\rightarrow_R^* \subseteq \Rightarrow_R^* \subseteq (\rightarrow_R^*)^* = \rightarrow_R^*$. $\square$

Definition (*complete development*). For R orthogonal, define $^\circ : \mathcal{T}(\Sigma, \mathcal{X}) \rightarrow \mathcal{T}(\Sigma, \mathcal{X})$ by well-founded recursion on $\sqsubset$:

$$t^\circ = \begin{cases} t & \text{if } t \in \mathcal{X}, \\ r\sigma^\circ & \text{if } t = l\sigma \text{ is a redex at the root, } \sigma^\circ(x) := (x\sigma)^\circ, \\ f(t_1^\circ, \dots, t_k^\circ) & \text{if } t = f(t_1, \dots, t_k) \text{ is not itself a redex.} \end{cases}$$

Intuitively, t° is obtained by contracting *every* redex of t simultaneously (and every redex thereby created inside the substituted variable instances, recursively).

Lemma (*strip lemma*). If $t \Rightarrow_R t'$, then $t' \Rightarrow_R t^\circ$.

Proof. By induction on $\sqsubset$. If t is not itself a redex and $\varepsilon \notin P$: partition P by argument and apply the induction hypothesis argument-wise. If $t = l\sigma$ is a redex and $\varepsilon \notin P$: non-overlapping forces every position of P to lie inside some variable instance $x\sigma$ (left-linearity making these instances disjoint); apply the induction hypothesis to each, then transport through the fixed skeleton l . If $\varepsilon \in P$: since P is an antichain, $P = \{\varepsilon\}$, and $t' = r\sigma$; joining each variable instance $x\sigma \Rightarrow_R (x\sigma)^\circ$ (trivially, via the induction hypothesis applied to the empty step) and transporting through r gives $r\sigma \Rightarrow_R r\sigma^\circ = t^\circ$. $\square$

Lemma (*diamond property for $\Rightarrow_R$*). If $t \Rightarrow_R t_1$ and $t \Rightarrow_R t_2$, there exists w with $t_1 \Rightarrow_R w$ and $t_2 \Rightarrow_R w$.

Proof. By the strip lemma applied to each step: $t_1 \Rightarrow_R t^\circ$, $t_2 \Rightarrow_R t^\circ$. Take $w := t^\circ$. $\square$

Theorem (*Orthogonal confluence (Rosen)*). Every orthogonal TRS is confluent.

Proof. By $\Rightarrow_R^* = \rightarrow_R^*$, it suffices to show $\Rightarrow_R$ is confluent. Given $t = u_0 \Rightarrow_R \dots \Rightarrow_R u_m = t_1$ and $t = v_0 \Rightarrow_R \dots \Rightarrow_R v_k = t_2$, we show by induction on $m + k$ that u_m, v_k are joinable. If $m = 0$ or $k = 0$, one sequence is trivial. Otherwise, the diamond property applied to $u_0 \Rightarrow_R u_1$ and $v_0 \Rightarrow_R v_1$ gives z with $u_1 \Rightarrow_R z$, $v_1 \Rightarrow_R z$; applying the induction hypothesis twice (to $t_1 \xleftarrow{*} u_1 \Rightarrow_R z$, then to $w_1 \xleftarrow{*} z \Rightarrow_R^* t_2$) closes the induction. $\square$

Remark. Note the contrast with the theorem on the union of confluent TRS: orthogonality requires no disjointness of signatures and no termination assumption whatsoever, but instead requires the syntactic conditions of non-overlap and left-linearity *within a single* system. The two results address genuinely different questions and are not special cases of one another.

8 Undecidability via Turing Machines

This section is independent from the rest of the paper: it illustrates, via an encoding of Turing machines as term rewriting systems, that termination of an arbitrary TRS is undecidable. It may be skipped by a reader only interested in the confluence-theoretic development of the preceding sections.

Definition (*Turing machine*). A deterministic Turing machine is a tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$, where Q is a finite set of states, Σ the finite input alphabet (with blank $\sqcup \notin \Sigma$), $\Gamma \supseteq \Sigma \cup \{\sqcup\}$ the tape alphabet, $q_0 \in Q$ the initial state, $q_{\text{acc}} \neq q_{\text{rej}} \in Q$ the accepting/rejecting states, and $\delta : (Q \setminus \{q_{\text{acc}}, q_{\text{rej}}\}) \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$ the transition function.

Remark. We restrict attention to deterministic Turing machines throughout this work, since this is the setting needed for the reduction to term rewriting and for the undecidability results that follow. The non-deterministic case is not treated here.

8.1 Encoding the tape

We assume three distinct symbols $0, 1, \sqcup$ denoting a value of 0, 1, or blank. A fourth symbol $\sqcup_!$ denotes an infinite sequence of blanks. Denoting $c(x, l')$ a sequence starting with x and continuing with l , the tape is modeled as $\text{tape}(l, c, r)$, with l, r the sequences to the left/right and $c \in \{0, 1, \sqcup\}$ the current symbol; both l, r are either $\sqcup_!$ or of the form $c(x, y)$. This representation is not unique; the *band normalisation* rule

$$c(\sqcup, \sqcup_!) \rightarrow \sqcup_!$$

removes unnecessary blanks. The moves are modeled by

$$\text{left}(\text{tape}(c(x, y), x', y')) \rightarrow \text{tape}(y, x, c(x', y')), \quad \text{right}(\text{tape}(x, y, c(x', y'))) \rightarrow \text{tape}(c(x, y), x', y').$$

There are two critical pairs between each move rule and band normalisation; orienting them via Knuth–Bendix gives two additional rules, yielding a convergent system R_{tape} .

Proposition (*termination of the band normalisation rule*). $R_{\sqcup} = \{c(\sqcup, \sqcup_!) \rightarrow \sqcup_!\}$ is terminating.

Proof. Each step replaces a subterm of 3 function symbols by one of 1: the total symbol count strictly decreases at each step, and cannot decrease infinitely often. $\square$

Proposition (*confluence of the band normalisation rule*). $R_{\sqcup}$ is confluent.

Proof. By Newman’s Lemma, it suffices to show local confluence. If two steps occur at the same position, they yield the same term. If at different positions $p \neq q$, they cannot be nested (the term $c(\sqcup, \sqcup_!)$ has no strict subterm equal to itself), so p, q are disjoint and the steps commute. $\square$

8.2 Encoding configurations and transitions

Definition (*input term*). The set I_{Σ} of input terms is defined by: $\sqcup_! \in I_{\Sigma}$; if $w \in I_{\Sigma}$, $a \in \Sigma$, then $c(a, w) \in I_{\Sigma}$.

Definition (*configuration term*). A *configuration* of M is a term $\text{conf}(q, \text{tape}(l, c, r))$, $q \in Q$. Given an input $w = c(a_1, \dots, c(a_n, \sqcup_!) \dots)$, the *initial configuration* is $t_{M, w} = \text{conf}(q_0, \text{tape}(\sqcup_!, a_1, c(a_2, \dots)))$.

Definition (*transition rules*). For every (q, a) with $\delta(q, a) = (q', a', d)$, R_M contains (for $d = R$) $\text{conf}(q, \text{tape}(l, a, r)) \rightarrow \text{right}(\text{conf}(q', \text{tape}(l, a', r)))$, or (for $d = L$) the symmetric rule with left. We further add wrapper rules pulling $\text{right}(\text{conf}(q, T)) \rightarrow \text{conf}(q, \text{right}(T))$ (and symmetrically for left), so the tape-move rules act on the tape component. R_M is the union of $R_\square$, the tape-move rules, the wrapper rules, and the transition rules.

Definition (*halting configuration*). $\text{conf}(q, \text{tape}(l, c, r))$ is *halting* if $q \in \{q_{\text{acc}}, q_{\text{rej}}\}$.

Lemma (*state and current symbol are unaffected by $R_\square$*). A single band normalisation step from $\text{conf}(q, \text{tape}(l, c, r))$ changes neither q nor c .

Proof. The rule's left-hand side is rooted by the list constructor c , distinct from conf and from any single element of Q or Γ ; the redex position must therefore lie strictly inside l or r . $\square$

Definition (*computation of M*). By induction on $k \geq 0$: $C_0 = t_{M,w}$; as long as C_k is not halting, with $\delta(q, c) = (q', c', d)$, $C_{k+1} = \text{conf}(q', T^*)$, T^* the R_{tape} -normal form of the tape after the move. M *halts* on w if some C_n is halting.

Lemma (*transition subsequence*). Along a finite reduction $t_{M,w} = D_0 \rightarrow_{R_M} \cdots \rightarrow_{R_M} D_N$, at each transition step index i_j , the configuration D_{i_j+1} has the same state and current symbol as C_j , and $D_{i_j+1} \rightarrow_{R_{\text{tape}}}^* C_j$.

Proof. By induction on j , using Lemma [state and current symbol are unaffected by $R_\square$] and confluence/termination of R_{tape} to track the unique normal form of the tape through interleaved $R_\square$ -steps. $\square$

Proposition (*Turing machine by a TRS*). M halts on w iff $t_{M,w}$ reduces, in R_M , to an irreducible term.

Proof. ($\Rightarrow$) Concatenate the n blocks of steps (one transition plus finitely many $R_\square$ -steps) corresponding to each computation step; the final halting configuration is irreducible, since no transition rule matches an accepting/rejecting state, and its tape is already normal. ($\Leftarrow$) Given an irreducible D_N , Lemma [transition subsequence] identifies its state with that of some C_m ; irreducibility forces $q_m \in \{q_{\text{acc}}, q_{\text{rej}}\}$, so C_m is halting. $\square$

Lemma (*reduction of TRS termination to TM halting*). R_M is terminating iff M halts from every configuration.

Proof. ($\Rightarrow$) Every reduction from a configuration C is finite, hence reaches an irreducible term, i.e. M halts from C (Proposition [Turing machine by a TRS]). ($\Leftarrow$) Every reduction sequence of R_M either progresses a configuration (finitely many steps, by the above) or normalises the tape (finitely many, by termination of $R_\square$). $\square$

8.3 Undecidability results

Definition (*encoding of Turing machines*). We fix an effective encoding $\langle M \rangle \in \{0,1\}^*$ of any deterministic Turing machine, identifying *algorithm* with *always-halting Turing machine*.

Theorem (*Halting problem for Turing machines*). There is no algorithm that, given an arbitrary deterministic Turing machine M , always terminates and correctly decides whether M halts on $\langle M \rangle$.

Proof. Suppose such H exists. Define D : on input $\langle M \rangle$, compute $H(\langle M \rangle)$; if "yes", loop forever; if "no", halt. Running D on $\langle D \rangle$: if D halts, $H(\langle D \rangle) = \text{"yes"}$, forcing D to loop — contradiction. If D does not halt, $H(\langle D \rangle) = \text{"no"}$, forcing D to halt — contradiction. $\square$

Theorem (*Halting problem for TRS*). There is no algorithm that, given an arbitrary TRS R , always terminates and correctly decides whether R is terminating.

Proof. Given such an algorithm A , and an arbitrary M , build R_M ; running A on R_M would decide, by Lemma [reduction of TRS termination to TM halting], whether M halts from every configuration, in particular on $\langle M \rangle$ — contradicting the previous theorem. $\square$

Conclusion

We have surveyed the main pillars of equational reasoning via term rewriting: the notion of confluence and its two classical characterisation tools (Newman's Lemma, Section 3; the Critical Pair Theorem, Section 5, resting on the unification algorithm of Section 4); the modular behaviour of these properties under disjoint unions (Section 7), where confluence is modular but termination is not, unless left-linearity is assumed; and the principal algorithmic technique for restoring confluence, Knuth–Bendix completion (Section 6). Section 8 situates these results against the fundamental limit imposed by the undecidability of termination itself. Together, these results give a coherent picture of when, and how, rewriting can be turned into a decision procedure for equational validity — and of the genuine limits of that programme.

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